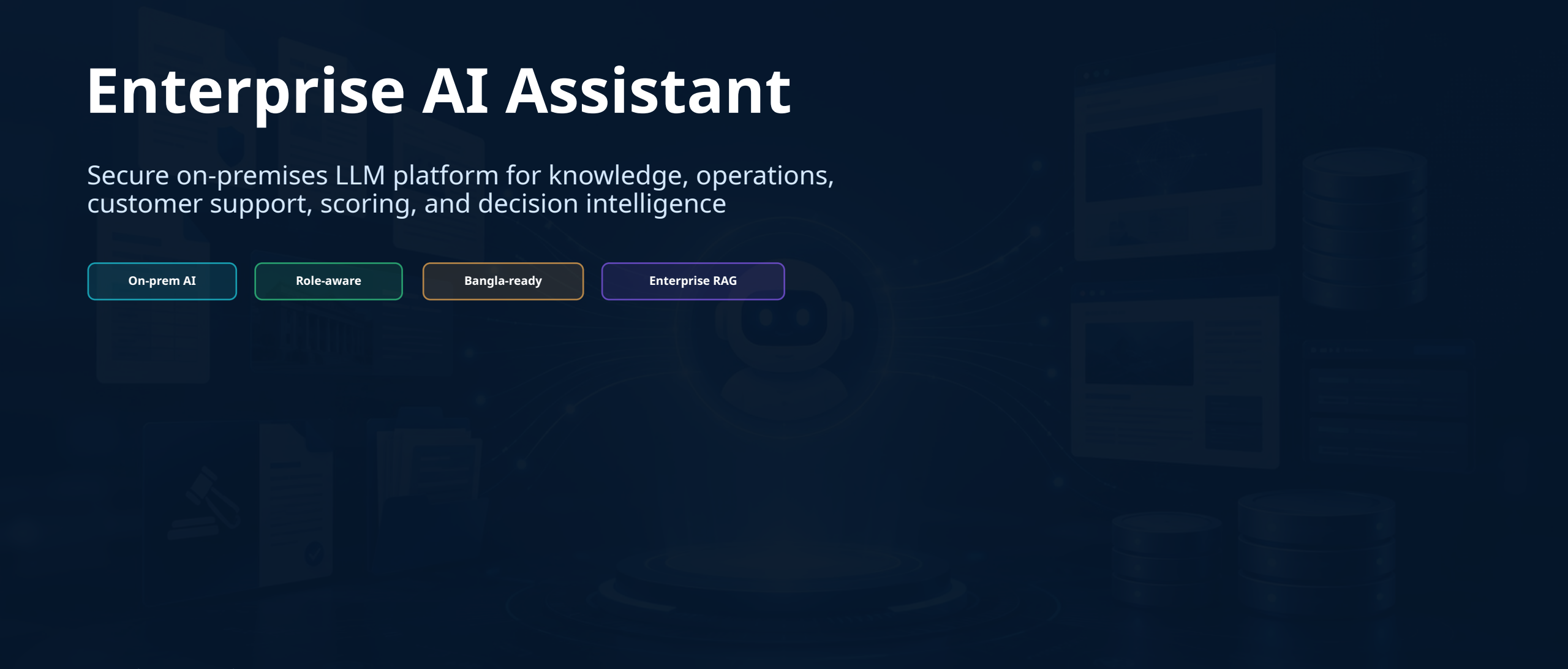
MANAGEMENT BRIEFING

Enterprise AI Assistant

Secure on-premises LLM platform for knowledge, operations, customer support, scoring, and decision intelligence

On-prem AI | Role-aware | Bangla-ready | Enterprise RAG



Internal presentation | May 2026

Executive case for management

What the platform changes for the organization

The solution turns scattered institutional knowledge and operational systems into a governed AI interface that answers, routes, searches, scores, and explains in plain language.

Business outcome

Faster staff response, improved decision support, repeatable service quality, and stronger institutional memory.

Control outcome

Role-based access, internal hosting, audit trail, and clear separation between public and sensitive modules.

Designed for internal reality

Works with Oracle, NiFi, Qdrant, Ollama, Kubernetes, NFS, documents, notices, policies, and website content.

Bangla-first adoption

Supports Bangla and English user queries, with a path to Bangla voice input for field and branch users.

Action-oriented AI

Not only chat: it can trigger RAG search, Oracle-backed APIs, Zypl scoring, live logs, and admin workflows.

1

Secure interface for multiple departments

24/7

Knowledge access without waiting for human routing

On-prem

No public cloud dependency for core LLM

RBAC

Controlled access by role/module

The operational problem we are solving

Institutional data exists, but staff access is fragmented

Document overload

Policies, circulars, notices, manuals, and uploaded files are difficult to search consistently.

System fragmentation

Oracle, core loan data, cashflow data, website information, and external scoring are not available through one interface.

Manual dependency

Users rely on specific technical people or department experts to interpret data and documents.

Language barrier

Bangla-first users need natural language access, not only dashboards or SQL screens.

Governance risk

Generic AI tools cannot be used safely for sensitive internal data without access control and isolation.

Management need: a secure enterprise AI layer that reduces friction without compromising control.

Strategic vision

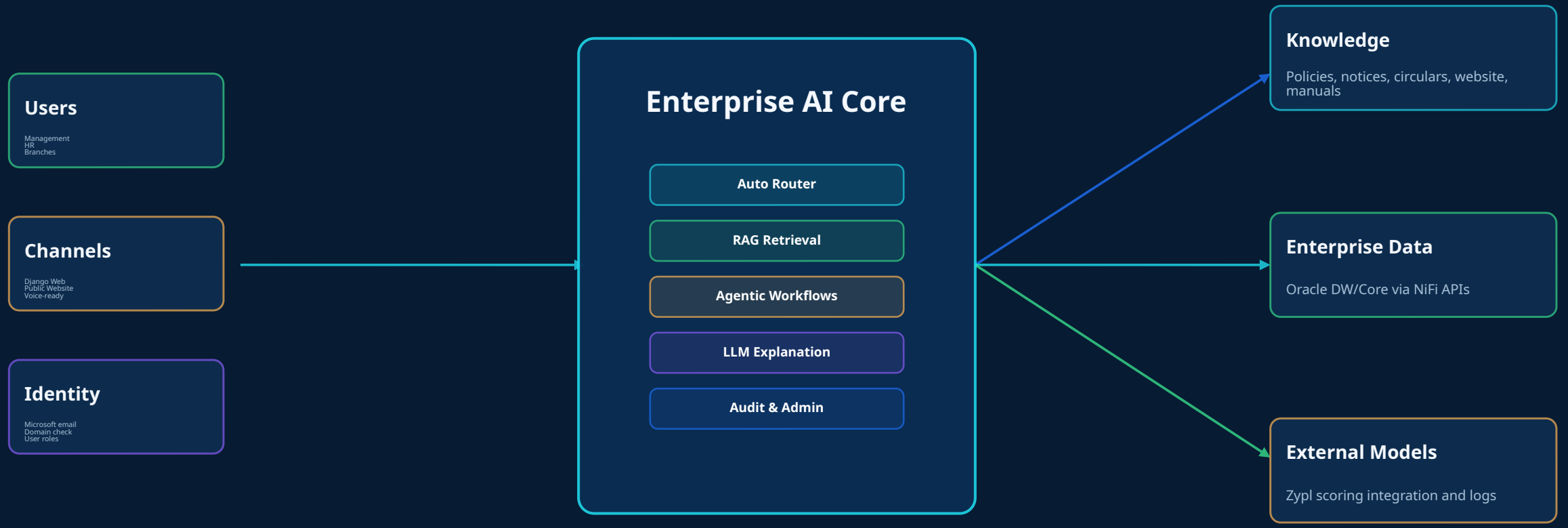
A central AI operating layer for the whole organization

From information search to intelligent action



Solution at a glance

A secured AI chat layer over knowledge, data, models, and workflows



Core principle: sensitive data stays governed; answers are grounded in approved sources and live systems.

Stakeholder value map

Different users get different value through the same platform

Management

Decision intelligence, rollout visibility, adoption metrics, risk controls

HR & Admin

Policy Q&A, notices, circular upload, HR knowledge retrieval

Branch staff

Customer support, loanee lookup, branch-wise queries, Bangla explanations

Cashflow users

Agent-guided scoring, Zypl API integration, live scoring logs

IT & Data team

On-prem control, K8s deployment, Qdrant tools, monitoring, data pipelines

Public users

Website Q&A without exposing internal modules or requiring login

One platform, many role-based experiences - no single user sees everything.

Capability portfolio

What the platform can do today and what it is prepared to do next

AI Chat + Auto Router

Natural language question goes to correct module or workflow.

RAG Knowledge Search

Grounded answers from policy, notice, circular, website, and curated data collections.

Oracle/NiFi Live Data

Numbers and operational facts retrieved from internal databases, not hallucinated.

Zypl Scoring Agent

Collects customer features, sends API request, returns score and logs.

Admin Console

Upload, index, monitor collections, sync website, manage module tools.

Bangla + Voice Path

Bangla-first user journey with future offline/controlled speech-to-text.

Security & RBAC

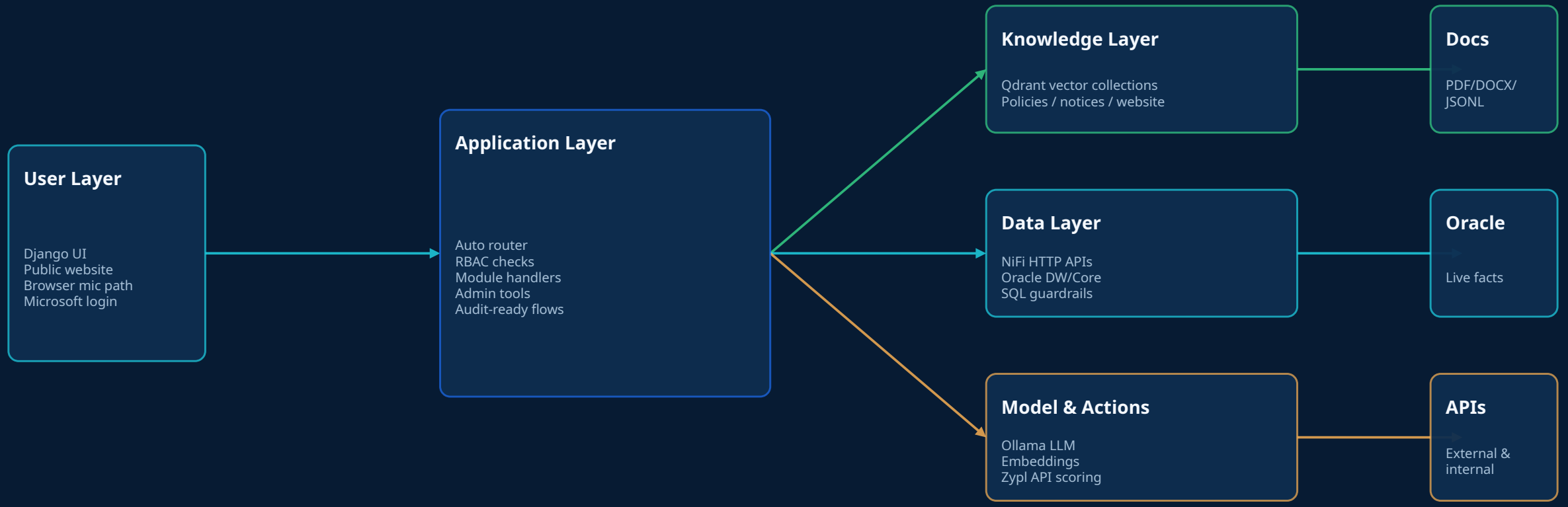
User roles restrict modules, sensitive flows, and admin actions.

Monitoring

Prometheus/Grafana/K8s health for production operations.

High-level architecture

Modular, governed, and extensible by design



Production foundation: Kubernetes, NFS persistent volumes, NodePort/internal DNS, Prometheus/Grafana, and secure network boundaries.

RAG knowledge lifecycle

How approved documents become reliable AI answers



Policies, notices, circulars, website pages, manuals, JSONL datasets

Text parsing, cleanup, metadata capture, language fields

Topic-wise or section-wise chunks with clause/date/source tags

nomic-embed-text vectors; cosine search in Qdrant

Top matches retrieved by module and access context

Aya model explains using grounded excerpts and safe fallback

✓ Governance rule

Only approved collections should answer sensitive internal questions.

! Quality rule

When no clear excerpt exists, the assistant should say so instead of inventing.

↻ Update rule

Admin teams can upload and reindex notices/circulars continuously.

Live enterprise data integration

Numbers come from internal systems, not from the language model



API

Controlled access path

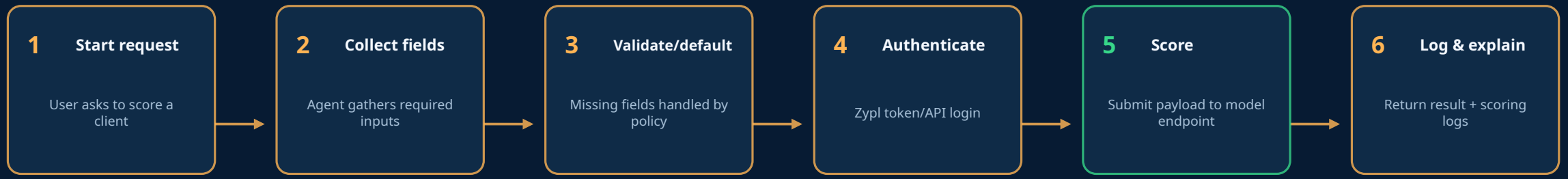
Design guardrail: the LLM explains results, but transactional facts and numeric answers are retrieved from Oracle through controlled NiFi flows.

SQL

Guarded execution

Agentic cashflow scoring flow

From conversational data collection to API-backed risk score



Current strength

The assistant can collect client information, call Zypl securely, and return a live scoring result inside chat.

Batch direction

Excel upload and batch scoring can become a controlled back-office workflow with result export.

Governance requirement

Cashflow role must control access; logs should be visible only to authorized users.

Security and access governance

Built for controlled internal usage, not open-ended public AI

Identity

Microsoft email login, allowed domain, auto-create default general user.

RBAC

Module-level access: general, HR, policy, notice, cashflow, core, customer support, admin.

Data boundary

On-prem model serving; sensitive data remains inside controlled infrastructure.

Public isolation

Website Q&A can be open without exposing internal modules.

Audit readiness

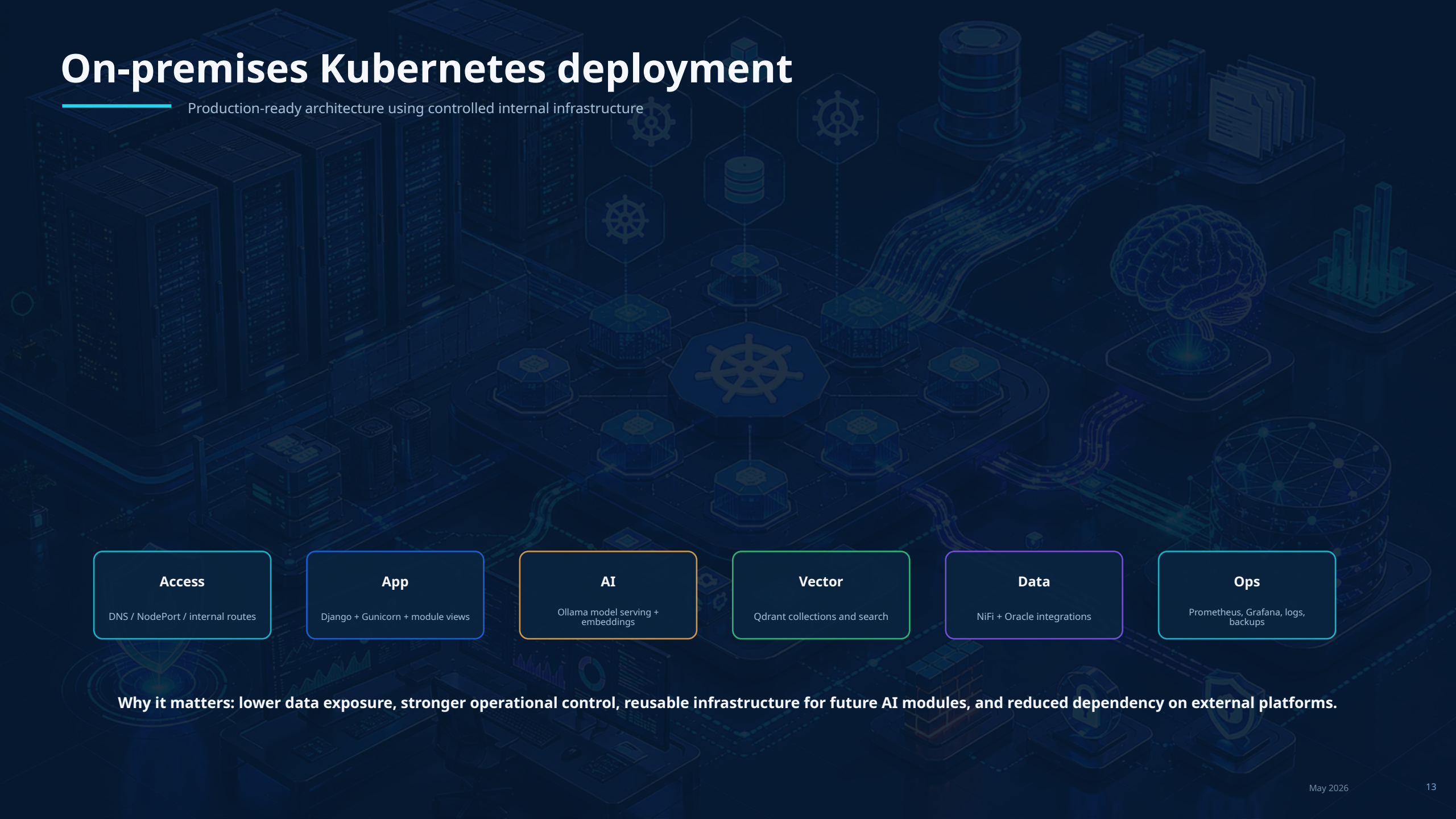
Logs, scoring history, admin actions, and system health can be monitored.

Least privilege

Users only access workflows needed for their role.

On-premises Kubernetes deployment

Production-ready architecture using controlled internal infrastructure



Access

DNS / NodePort / internal routes

App

Django + Gunicorn + module views

AI

Ollama model serving + embeddings

Vector

Qdrant collections and search

Data

NiFi + Oracle integrations

Ops

Prometheus, Grafana, logs, backups

Why it matters: lower data exposure, stronger operational control, reusable infrastructure for future AI modules, and reduced dependency on external platforms.

AI, model, and data layer

Separation of reasoning, retrieval, and system-of-record facts

Generation model	**Aya via Ollama**	Bangla/English explanation, summarization, controlled chat responses
Embedding model	**nomic-embed-text**	Creates 768-dimensional vectors for Qdrant semantic search
Vector database	**Qdrant**	Collections for HR notices, policies, website, cashflow, core/customer support
Workflow engine	**Apache NiFi**	Controlled HTTP APIs for Oracle queries and integration steps
System of record	**Oracle DW/Core**	Authoritative operational facts, loan/customer metrics, transaction data
Persistence	**NFS volumes**	Model cache, document storage, service data, backups

Knowledge modules: HR, policy, and notices

Turning internal documents into governed Q&A and self-service support

HR

HR Policy RAG

Curated policy JSONL imports, topic metadata, clause ranges, Bangla/English labels, and grounded answers.

N

Notice RAG

Document upload, notice metadata, Qdrant indexing, list/detail/search/reindex routes, and role-based retrieval.

Policy Admin

Admin tools to browse Qdrant collections, upload JSONL, manage topics, and restore/copy collections.

User experience example

“What is the leave policy?” or “Show recent circular about HR benefits” → router selects HR/notice collection → retrieves approved source → assistant answers in Bangla or English.

Upload

Admin-controlled ingestion

Index

Vector search in Qdrant

Answer

Grounded by approved source

Reindex

Continuous updates

Operational modules: core, loan, and customer support

From static Q&A to live operational intelligence

Core/Loan queries

Branch code, loanee number, balance, paid amount, transaction date, and other operational data via Oracle/NiFi.

Customer support knowledge

Reusable support answers, customer-facing guidance, ticket creation/status paths for authorized roles.

Sensitive data rules

Branch staff, support staff, managers, and admin roles receive different capabilities.

SQL guardrails

Intent extraction and templates reduce unsafe free-form database queries.

Strategic value: branch and support teams can ask business questions in natural language while the system keeps authoritative data in Oracle.

Public website RAG module

Approved public Q&A without exposing internal systems



Why it matters

Reduces repetitive website/program questions and improves access for external stakeholders.

Isolation principle

Public website questions must never route to HR, core, cashflow, or customer-sensitive modules.

Admin control

Website sync and collection health should be visible from the admin console.

Bangla-first user experience and voice path

Adoption improves when staff can use natural Bangla and simple voice input

বাংলা প্রশ্ন → intent detection → secure module routing → grounded answer

Current direction

Bangla and English questions through the Django chat UI, with model-side Bangla explanation using Aya.

Offline STT option

faster-whisper/whisper-server can provide controlled internal speech-to-text for Bangla users.

Production requirement

Better microphone capture, model selection, GPU/CPU planning, and browser permission handling.

Management message: Bangla support is not optional; it is the primary adoption lever for branch, field, and non-technical users.

Admin console and operating tools

Management control depends on visible administration and repeatable operations

Q Qdrant Tools

Browse collections, copy/restore, switch base URL, validate vector counts

↑ Upload JSONL

Curated policy/topic ingestion for controlled knowledge updates

N HR Notice Admin

Upload, list, detail, search, and reindex notices/circulars

↻ Website Sync

Admin-triggered clean sync, sync-only, and scheduled daily refresh

RB RBAC Admin

Create users, assign role/module access, separate sensitive functions

Z Zypl Logs

View scoring history and troubleshoot batch/API flows

Admin tools are the difference between a prototype and a maintainable enterprise platform.

Observability and reliability model

Operational confidence for production-grade AI services

Service health

- ✓ Django / Gunicorn
- ✓ Qdrant endpoints
- ✓ Ollama generation
- ✓ NiFi flows
- ✓ Oracle availability
- ✓ Kubernetes pods

Production metrics

- Question volume
- Latency by module
- Error rate
- Retrieval hit quality
- Scoring throughput
- User adoption

Runbook controls

- → Backups / NFS data
- → Model cache checks
- → Collection restore
- → Fallback endpoints
- → Restart procedures
- → Incident review

Expected business benefits

A practical value model for management

Speed

Faster answers to policy, notice, customer, and data questions

Consistency

Same approved answer logic across branches and departments

Control

Internal hosting, RBAC, admin tools, audit-ready logs

Scale

Reusable platform for new modules, data sources, and workflows

✓ Knowledge productivity

Less time searching documents; faster onboarding for new staff.

✓ Decision support

Live Oracle data plus AI explanation improves clarity for operational questions.

✓ Service quality

Support staff can respond faster with approved knowledge and live context.

✓ Risk management

Sensitive workflows are gated by role and internal infrastructure.

✓ Innovation foundation

Agentic workflows can be added for scoring, logs, tickets, reports, and analytics.

Risk controls and mitigation plan

Known risks are manageable when designed into operations

Hallucination risk

Ground answers in Qdrant excerpts; say “not found” when source is unclear.

Access leakage

Enforce RBAC before retrieval/action; public website module isolated.

Model latency

Monitor Ollama timeouts, tune model size, add health checks and fallback messages.

Data correctness

Use Oracle/NiFi for facts and numbers; LLM only explains retrieved results.

Document quality

Curate metadata, topic IDs, dates, language, and approval status before indexing.

Operational dependency

Create admin tools, runbooks, backups, and service ownership model.

Implementation maturity

From prototype capabilities to enterprise operating model



Foundation

On-prem stack: Django, Ollama, Qdrant, NiFi, Oracle, K8s, NFS

Module delivery

HR, policy, notice, website RAG, core data, cashflow scoring

Governance

RBAC, admin console, collection management, secure routing

Production hardening

Monitoring, runbooks, backups, latency controls, usage analytics

Scale-out

More departments, report generation, ticketing, voice, advanced workflows

Current position: strong technical foundation and working modules; next phase is standardization, monitoring, governance, and adoption.

Roadmap to enterprise rollout

Phased plan to move from working solution to managed platform

0-30 days

Stabilize

Finalize RBAC, admin links, known module routing, backup scripts, and production runbook.

31-90 days

Pilot

Run pilot with HR, cashflow, support, and core data teams; capture feedback and metrics.

3-6 months

Scale

Add dashboards, batch scoring, voice text input, improved RAG quality checks, and adoption training.

6-12 months

Institutionalize

Create AI governance committee, service ownership, model lifecycle, and new module pipeline.

Roadmap principle: stabilize first, then pilot with measurable adoption, then scale modules under governance.

Management decisions required

What management support is needed to move forward

1 Approve platform direction

Recognize the enterprise AI assistant as an internal digital capability, not an isolated experiment.

2 Assign module owners

Each business module needs a responsible department owner for content quality and approval.

3 Authorize pilot users

Select controlled pilot groups from HR, support, cashflow, core data, and management.

4 Define governance

Create data sensitivity rules, acceptable use, access approval, and escalation process.

5 Resource production hardening

Allocate time for monitoring, backup, security review, GPU/CPU planning, and documentation.

6 Track value metrics

Measure adoption, time saved, query volume, latency, answer quality, and error reduction.

Decision request: approve a formal pilot and governance track for the next 90 days.

Suggested success metrics

How to prove value after pilot launch

A Adoption

Monthly active users by department and role

T Productivity

Average time saved per policy/data/support query

Q Quality

Answer helpfulness rating and retrieval evidence hit rate

R Reliability

Module latency, errors, timeouts, uptime

G Governance

Access violations blocked, audit completeness, role review cycle

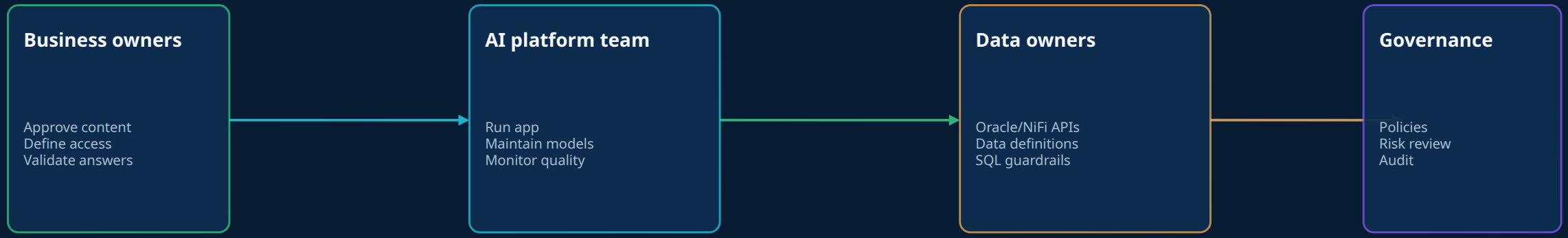
B Business usage

Scoring requests, notices indexed, customer queries resolved

Recommended pilot dashboard: adoption + latency + answer quality + module usage + incident log.

Target operating model

Business ownership plus IT platform ownership



Content lifecycle

Submit → approve → index → test → publish → monitor.

Access lifecycle

Request → approve → assign role → review periodically.

Model lifecycle

Evaluate → deploy → monitor → update → rollback if needed.

CLOSING MESSAGE

A secure AI operating layer for the organization

The platform is ready to move from technical success to managed business adoption: governed, measurable, and expandable.

1 **Approve pilot**

90-day controlled rollout with selected users.

2 **Assign owners**

Business and IT owners for each module.

3 **Measure value**

Track adoption, accuracy, latency, and impact.

Thank you

Appendix: technical inventory

Current platform components and example collections

Application	Django app, module routers, admin URLs, role checks
LLM	Aya via Ollama, dynamic model configuration
Embedding	nomic-embed-text, 768-dim vectors
Vector DB	Qdrant with collections: hr_notices, buro_policy, buro_website, dw_cashflow, buro_core, etc.
Integration	NiFi HTTP APIs, Oracle 11g/DW/Core, Zypl API
Infrastructure	On-prem Kubernetes, NFS, NodePort/internal DNS, Prometheus/Grafana

Note: inventory should be validated against production before external circulation.